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Sport-Specific Differences in Vertical Jump Force-Time Metrics Between Professional Female Volleyball, Basketball, and Handball Players

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Abstract

Cabarkapa, DV, Cabarkapa, D, Aleksic, J, and Fry, AC. Sport-specific differences in vertical jump force-time metrics between professional female volleyball, basketball, and handball players. *J Strength Cond Res* 39(5): 587–592, 2025—The purpose of the present study was to examine the sport-specific differences in countermovement vertical jump (CMJ) force-time metrics among professional female volleyball, basketball, and handball players. Ninety-four athletes volunteered to participate in the present study (i.e., 41 volleyball, 20 basketball, and 33 handball). After a brief warm-up procedure, each athlete performed 3 nonconsecutive CMJs while standing on a force plate system sampling at 1,000 Hz. Nineteen force-time metrics were selected for performance analysis purposes, including both eccentric and concentric phases of the jumping motion. A one-way analysis of variance with Tukey post-hoc comparisons was used to examine statistically significant differences in each dependent variable across 3 sports ($p < 0.05$). The results reveal that volleyball athletes demonstrate significantly greater impulse, velocity, and mean and peak power during the eccentric phase of the CMJ compared with both basketball and handball players, and longer eccentric duration than basketball players. During the concentric phase, volleyball athletes showed significantly greater duration, impulse, and velocity compared to their handball and basketball counterparts, with higher mean and peak force observed only in comparison with basketball players. In addition, volleyball athletes had significantly greater jump height and deeper countermovement depth than the other 2 groups. However, the difference in reactive strength index-modified was detected only between the handball and volleyball athletes, with volleyball players exhibiting greater values. Overall, these findings can help sports practitioners with the development of specialized performance-enhancement training programs for athletes competing in sports such as basketball, volleyball, and handball.

Key Words: eccentric, concentric, force, power, impulse, development, monitoring, team sports

Introduction

Lower-body strength and power are crucial attributes for athletes competing in sports that require rapid change of direction and jumping movements, such as volleyball, basketball, and handball (20,27,34). However, despite being some of the most popular sports in the world for women's participation and the ones that attract a large cohort of spectators, these performance characteristics remain underexplored in the scientific literature, especially when solely focused on examining female athletes. Studies have shown that competitive volleyball players typically perform 123–180 jumps per game for maneuvers such as hitting and blocking (20), whereas basketball players generally execute 50 jumps during a game (34). Moreover, many sport-specific actions, such as rebounding and blocking in the game of basketball (6) or dodging in handball (35), require players to constantly perform multidirectional movements (6) and engage in high-intensity activities across short distances (23,30). Thus, the collective insights from the aforementioned studies underscore the pivotal role of lower-body strength and power in attaining peak performance potential in these sports and highlight the need for assessing these attributes in a competitive framework (15).

The countermovement vertical jump (CMJ) is widely implemented by sports scientists and strength and conditioning practitioners as a standard testing modality for the assessment of lower-body strength and power characteristics (8). Force plates, recognized as the "gold standard" measuring instruments, are frequently used for CMJ assessment within sports-specific contexts (9,28). This is largely because of the noninvasive and time-efficient nature of the testing procedure, which is essential in professional sports settings (8). More importantly, it is because of their capability to rapidly produce a wide range of force-time metrics that allow for the detailed analysis of athletes' performance (2,21). Based on the nature of the assessment, CMJ force-time metrics have been previously used in volleyball (12), basketball (10,29), and handball (27) to assess athlete's overall neuromuscular capabilities (29), assess injury risk (17), monitor the influence of fatigue (10,12,28), and distinguish players at different playing positions or level of play (8).

To date, several studies have compared the sport-specific differences between volleyball, basketball, and handball players (3,15,26), collectively highlighting the unique demands of each sport. In terms of anthropometric characteristics, basketball

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players were found to be taller on average compared with their volleyball and handball counterparts, whereas volleyball and handball players demonstrated the lowest and highest body mass values, respectively (15,26). In addition, hand anthropometric measurements (i.e., hand size) and handgrip strength were significantly greater in handball players (3) than in basketball and volleyball athletes. In terms of lower-body muscle power, several studies pointed out superior vertical jump heights in volleyball players compared with the other 2 groups of athletes (5,26,33,37). On the other hand, basketball players revealed better agility performance (26) and reaction time tests (18), whereas handball players had superior acceleration speed (32) and handgrip strength (3).

Overall, these studies contribute to a broader understanding of the unique physical profiles associated with each sport, highlighting the diversity in training and performance requirements. However, with only a select number of studies focused on examining female athletes (4,36), the majority of research regarding CMJ performance solely involves male counterparts (5,17,33,37). Therefore, despite the critical role of vertical jump performance in sports such as volleyball, basketball, and handball, there is a lack of research focused on understanding the unique performance characteristics of female athletes, especially ones competing at a professional level of play.

Therefore, the aim of this study was to bridge the previously mentioned gap in the scientific literature and examine the sport-specific differences in CMJ force-time metrics among elite professional female volleyball, basketball, and handball players. The insights gained from this research may contribute to a more nuanced understanding of CMJ performance in female athletes and aid in the creation of more specialized training and performance-enhancement strategies for athletes competing in these sports.

Methods

Experimental Approach to the Problem

The present study used a cross-sectional study design to examine the sport-specific differences in lower-body neuromuscular performance characteristics among 3 groups of elite professional female athletes (i.e., volleyball [VB], basketball [BB], and handball [HB]).

Subjects

Ninety-four athletes volunteered to participate in the present study, from which 41 were volleyball (Mean ± SD; age = 25.3 ± 4.1 years; height = 180.0 ± 7.7 cm; body mass = 70.5 ± 7.6 kg), 20 basketball (age = 26.1 ± 3.3 years; height = 182.8 ± 7.2 cm; body mass = 68.8 ± 7.9 kg), and 33 handball (age = 24.5 ± 4.4 years; height = 177.8 ± 4.6 cm; body mass = 68.4 ± 10.7 kg) players. All athletes were active members of professional teams competing in a top-tier national league in Europe. The athletes included in the present investigation were free of musculoskeletal injuries that could limit CMJ performance and were cleared by their respective sports medicine staff to participate in team activities. The testing procedures performed in this investigation were previously approved by the University of Kansas Institutional Review Board and all subjects signed an informed consent document.

Procedures

The CMJ testing procedures were completed during the initial phase of the official competitive season period for each sport (i.e., volleyball, basketball, handball), 36–48 hours after the last practice session (e.g., rest period) to minimize the possible impact of fatigue. Upon arrival at the gym for the official team practice (i.e., 16:30–20:00 hours), athletes completed a standardized warm-up administered by their respective strength and conditioning coaches. The warm-up procedure was consistent across all 3 sports, and it consisted of a 5-minute low-intensity jog, followed by a set of dynamic stretching exercises (e.g., butt-kicks, A-skips, high-knees, lunge-and-twist, knee-to-chest). The overall duration of the warm-up protocol was 15 minutes. Then, each athlete stepped on a pair of portable uniaxial force plate (ForceDecks Max; VALD Performance, Brisbane, Australia) and performed 3 CMJs with no arm swing (i.e., hands on the hips during the entire movement). The force plate system was recalibrated/zeroed between each athlete. Throughout all testing procedures, research assistants provided a strong verbal encouragement to ensure that athletes provided maximal effort during each CMJ. Also, the athletes were instructed and reminded to focus on pushing the ground as forcefully as possible (19). If an athlete accidently used an arm swing, the jump trial was repeated and an average of 3 CMJ was used for performance analysis purposes.

Nineteen force-time metrics examined during both eccentric and concentric phases of the CMJ were included in the present investigation. The selection process was founded on previously published research reports, as variables commonly used in an applied sports setting and with an adequate level of validity and reliability (2,7,9,21). The force-time metrics of interest were eccentric and concentric peak and mean force and power, braking impulse, concentric impulse, eccentric and concentric duration, braking phase duration, contraction time, eccentric and concentric peak velocity, jump height (i.e., impulse-momentum calculation), reactive strength index (RSI)-modified (i.e., jump height divided by contraction time), and countermovement depth. Because of sport-specific differences, all variables dependent on the athlete's body mass (e.g., peak force, mean power) were expressed in relative terms. The contraction time started when the athlete's system mass was reduced by 20 N, which is termed the movement onset and ended at take-off. Similarly, the take-off was defined as the time point at which vertical force dropped below a threshold of 20 N. The eccentric phase was defined as the phase containing a negative center of mass velocity. The braking phase was defined as a subphase of the eccentric phase, starting at minimum force until the end of the eccentric phase. Impulse within each respective subphase was calculated as the area under the force-time curve. In addition, a detailed description of CMJ force-time metrics can be found in the VALD user manual (https://valdperformance.com/forcedecks/).

Statistical Analyses

Descriptive statistics were calculated for each dependent variable examined in the present investigation. Shapiro-Wilk test and Q-Q plots corroborated that the assumption of normality was not violated. A one-way analysis of variance with Tukey post-hoc adjustments was used to examine statistically significant differences in each force-time metric across 3 sports (i.e., volleyball, basketball, handball). Eta-squared (i.e., $\eta^2 < 0.01$-negligible; $0.01 \leq \eta^2 < 0.06$-small; $0.06 \leq \eta^2 < 0.14$-medium; $\eta^2 \geq 0.14$-large) and Cohen's d (i.e., difference between the 2 means divided by the pooled *SD*) were used to calculate the effect size magnitudes (i.e., $d = 0.2$-small, $d = 0.5$-moderate, $d = 0.8$-large) (9). Statistical significance was set a priori to $p \leq 0.05$. All statistical analyses were completed with SPSS (Version 26.0; IBM Corp., Armonk, NY), whereas graphical representations of the findings were generated by using *R* software (Version 4.2.1; *R* Foundation for Statistical Computing, Vienna, Austria).

Results

Between-group statistically significant differences were detected in eccentric braking impulse ($F_{[2,93]} = 19.227$; $p < 0.001$; $\eta^2 = 0.30$), eccentric duration ($F_{[2,93]} = 4.629$; $p = 0.012$; $\eta^2 = 0.09$), eccentric peak velocity ($F_{[2,93]} = 36.335$; $p < 0.001$; $\eta^2 = 0.44$), eccentric peak power ($F_{[2,93]} = 20.569$; $p < 0.001$; $\eta^2 = 0.31$), eccentric mean power ($F_{[2,93]} = 34.988$; $p < 0.001$; $\eta^2 = 0.43$), concentric duration ($F_{[2,93]} = 20.040$; $p < 0.001$; $\eta^2 = 0.32$), concentric impulse ($F_{[2,93]} = 14.505$; $p < 0.001$; $\eta^2 = 0.24$), concentric peak velocity ($F_{[2,93]} = 17.382$; $p < 0.001$; $\eta^2 = 0.28$), concentric peak force ($F_{[2,93]} = 5.637$; $p = 0.005$; $\eta^2 = 0.11$), concentric mean force ($F_{[2,93]} = 8.258$; $p < 0.001$; $\eta^2 = 0.15$), jump height ($F_{[2,93]} = 14.196$; $p < 0.001$; $\eta^2 = 0.24$), RSI-modified ($F_{[2,93]} = 5.055$; $p = 0.008$; $\eta^2 = 0.10$), and countermovement depth ($F_{[2,93]} = 44.167$; $p < 0.001$; $\eta^2 = 0.49$).

Volleyball players had significantly greater braking impulse (VB vs. BB $d = 1.06$; VB vs. HB $d = 1.40$), eccentric peak velocity (VB vs. BB $d = 1.61$; VB vs. HB $d = 1.87$), eccentric peak power (VB vs. BB $d = 1.13$; VB vs. HB $d = 1.42$), eccentric mean power (VB vs. BB $d = 1.54$; VB vs. HB $d = 1.85$), concentric impulse (VB vs. BB $d = 1.13$; VB vs. HB $d = 1.14$), concentric peak velocity (VB vs. BB $d = 1.19$; VB vs. HB $d = 1.26$), and jump height (VB vs. BB $d = 1.10$; VB vs. HB $d = 1.13$) when compared with both basketball and handball players (all $p < 0.001$). RSI-modified (VB vs. HB $d = 0.72$) was greater in volleyball when compared with handball players ($p < 0.001$). Eccentric duration (VB vs. BB $d = 0.85$; BB vs. HB $d = 0.65$) and concentric duration (VB vs. BB $d = 1.76$; VB vs. HB $d = 0.99$) were notably greater in volleyball players in comparison with basketball ($p = 0.012$ and $p = 0.037$) and handball players (both $p < 0.001$). Moreover, concentric peak force (VB vs. BB $d = 1.01$) and concentric mean force (VB vs. BB $d = 1.23$) were significantly greater in basketball when compared with volleyball players ($p = 0.003$ and $p < 0.001$), whereas countermovement depth (VB vs. BB $d = 2.05$; VB vs. HB $d = 1.91$) was lower in handball and basketball players than in volleyball players (both $p < 0.001$).

No statistically significant between-group differences were observed in braking phase duration ($F_{[2,93]} = 2.005$; $p = 0.141$; $\eta^2 = 0.04$), eccentric peak force ($F_{[2,93]} = 1.793$; $p = 0.172$; $\eta^2 = 0.04$), eccentric mean force ($F_{[2,93]} = 2.901$; $p = 0.060$; $\eta^2 = 0.03$), concentric peak power ($F_{[2,93]} = 2.866$; $p = 0.062$; $\eta^2 = 0.06$), concentric mean power ($F_{[2,93]} = 1.075$; $p = 0.346$; $\eta^2 = 0.02$), and concentric duration ($F_{[2,93]} = 2.748$; $p = 0.069$; $\eta^2 = 0.32$). See Figures 1–3 for illustrations of these results.

Discussion

The purpose of the present study was to examine the sport-specific differences in neuromuscular performance characteristics between elite professional female volleyball, basketball, and handball athletes. Volleyball athletes demonstrated significantly greater impulse, velocity, and mean and peak power during the eccentric phase of the CMJ compared with both basketball and handball players, and longer eccentric duration than basketball players. Also, during the concentric phase, volleyball athletes showed significantly greater duration, impulse, and velocity compared with their handball and basketball counterparts, with higher mean and peak force observed only in comparison with basketball players. Moreover, volleyball athletes achieved significantly greater jump height and deeper countermovement depth than the other 2 groups. However, the difference in RSI-modified was detected only between the handball

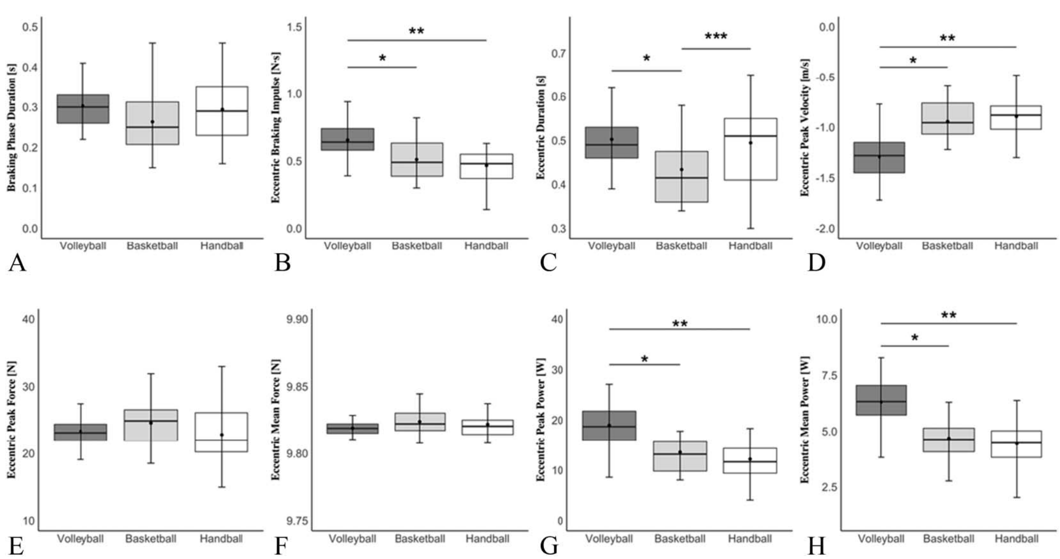

Figure 1. Between-group differences in force-time metrics within the eccentric phase of CMJ. *Statistically significant difference between volleyball and basketball. **Statistically significant difference between volleyball and handball. ***Statistically significant difference between basketball and handball. The black dot indicates the mean value and the black line the median value. The black dashes (whiskers) represent minimum and maximum and shaded areas interquartile ranges for each sport. CMJ = countermovement vertical jump.

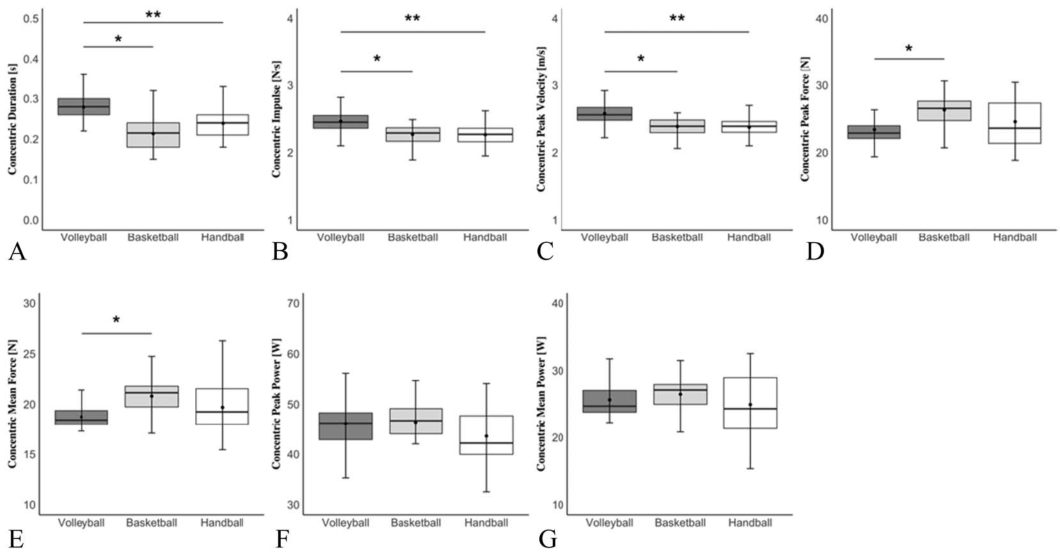

Figure 2. Between-group differences in force-time metrics within the concentric phase of CMJ. *Statistically significant difference between volleyball and basketball. **Statistically significant difference between volleyball and handball. The black dot indicates the mean value and the black line the median value. The black dashes (whiskers) represent minimum and maximum and shaded areas interquartile ranges for each sport. CMJ = countermovement vertical jump.

and volleyball athletes, with volleyball players exhibiting greater values. Finally, no significant differences were observed in any of the remaining force-time metrics of interest (i.e., braking phase duration, eccentric mean and peak force, and concentric mean and peak power).

Based on the findings of the present investigation, it can be concluded that the vertical jump performance considerably differs between basketball, volleyball, and handball athletes, potentially because of the unique movement patterns, and physical and physiological demands in these sports. In basketball, all players are required to be in a basic athletic "ready" stance, which allows them to maintain their balance, and quickly react to the explosive demands of the game such as shooting, passing, dribbling, and defending against an opponent (25). On the other hand, in volleyball, the "ready" position is determined by various factors such as the player's positioning in the front or back row, or on the offense or the defense at the beginning of the rally (14). Also, volleyball is a noncontact sport with no interference from the defender or opponent on the same side of the court, thus allowing athletes to have more time while executing skills such as attacking or blocking. This additional time during the vertical displacement for volleyball-specific skills may likely contribute to the significantly greater eccentric and concentric force-time metrics (e.g., duration, impulse, velocity) observed in the present investigation for volleyball players when compared with basketball athletes. However, considering the similarities between handball and basketball (e.g., high-intensity change of direction movements combined with physical contact and ball passing), it is understandable that handball players would demonstrate similar neuromuscular performance characteristics (e.g., eccentric force and power, concentric impulse, vertical jump height, RSI-modified) to the ones seen in basketball athletes.

Furthermore, volleyball players are required to perform a considerably higher volume of explosive countermovement jumps (e.g., blocking, attacking, serving) during the game than their basketball and handball counterparts, ultimately resulting in specific neuromuscular adaptations that enhance the efficiency of

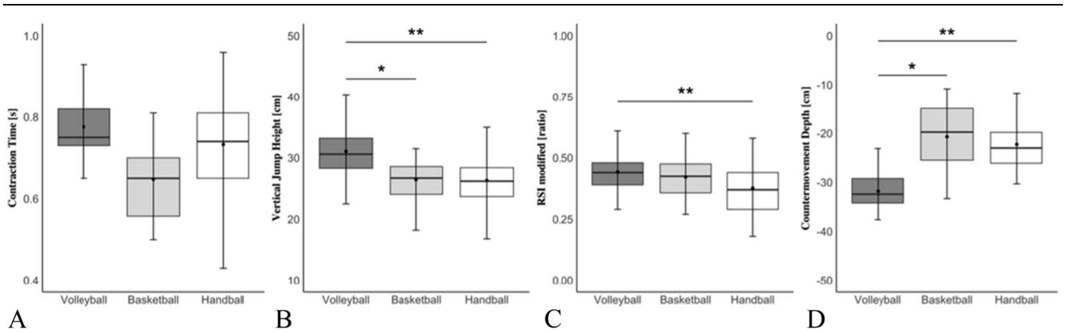

Figure 3. Between-group differences in CMJ outcome metrics. *Statistically significant difference between volleyball and basketball. **Statistically significant difference between volleyball and handball. The black dot indicates the mean value and the black line the median value. The black dashes (whiskers) represent minimum and maximum and shaded areas interquartile ranges for each sport. CMJ = countermovement vertical jump.

the athlete's stretch-shortening cycle (13,16,31). For example, during the training session, volleyball players perform approximately 120–180 jumps, which may differ based on their position on the team (11,20). On the other hand, both basketball and handball players seem to have a lower jump count (i.e., 49.8 ± 20.0 and 11.3 ± 7.2, respectively), but a significantly greater number of change of direction movements than volleyball athletes (24,34). Thus, the superior force-generating capabilities and greater vertical jump displacement observed in the present investigation for the cohort of volleyball athletes can be attributed to both the differences in external load demands between the aforementioned sports and the specificity of the CMJ test for this athletic population (i.e., volleyball) (1). However, this does not imply that the CMJ should be excluded from basketball and handball training/testing but rather highlights that volleyball players excel in this test because of direct alignment with their sport-specific demands (37).

Another important thing to consider when interpreting the results of the present study is the magnitude of the force-time metrics of interest and how they compare with different levels of competition (e.g., collegiate and lower professional levels). It is interesting to observe that volleyball players competing at the collegiate level of play exhibit notably lower eccentric and concentric impulse, force, and power, and vertical jump height and RSI-modified when compared with the professional volleyball athletes examined in the present investigation (13). For example, collegiate female volleyball players (i.e., National Association of Intercollegiate Athletics Division-I) demonstrated an eccentric impulse of ~0.45 $\text{Ns}\cdot\text{kg}^{-1}$ and eccentric power of ~10.5 $\text{W}\cdot\text{kg}^{-1}$, whereas professional volleyball athletes exhibited an eccentric impulse of ~0.6 $\text{Ns}\cdot\text{kg}^{-1}$ and eccentric power of ~ 19 $\text{W}\cdot\text{kg}^{-1}$ (13). In addition, based on the previous findings regarding the basketball athletic population (9,13) and the unpublished data collected in our laboratory on handball players competing at different levels of play in Europe, it can be implied that professional athletes in the previously mentioned sports tend to have considerably greater power-producing capabilities than their counterparts competing at lower-levels of play. Specifically, handball athletes examined in the present investigation demonstrated notably greater eccentric peak force (i.e., 23 vs. 10 $\text{N}\cdot\text{kg}^{-1}$) and power (i.e., 12 vs. 4 $\text{W}\cdot\text{kg}^{-1}$) than the athletes competing in the bottom-ranked team of European professional competition. Overall, it can be concluded that regardless of the sport (e.g., basketball, handball, volleyball), elite professional female athletes demonstrate significantly higher neuromuscular performance capabilities, as demonstrated during a CMJ test, than the players competing at the collegiate or lower professional levels of play.

Although these findings offer valuable insight into the CMJ performance characteristics of professional female athletes, this investigation is not without limitations. For instance, the athletes examined in the present investigation were tested at only one testing time point, which may not account for the detailed quantification of the external load that athletes were exposed to and variations in performance over different macro, meso, and micro training cycles during the competitive season. Also, the sample of subjects was homogenous as it included only female athletes competing at the professional level of play and it did not include analysis of potential performance differences in playing positions within each sport. Thus, future research should examine if these findings remain consistent across different competitive levels (e.g., youth, collegiate) and if they are sex-specific.

Practical Applications

The findings of the present study can help coaches, sports scientists, and strength and conditioning practitioners glean a deeper insight into sport-specific differences in CMJ performance characteristics that can be used for the development of specialized training and performance-enhancement strategies for athletes competing in team sports such as basketball, volleyball, and handball. This can ensure that each athlete's unique performance needs are met, leading to more effective training outcomes. Also, this sport-specific CMJ analysis approach is crucial for maximizing performance outcomes, as it can help with the identification of athletes' strengths (e.g., sufficient force production) and specific areas that need improvement (e.g., increasing the movement velocity to optimize power-producing capabilities). In addition, with force plates being commonly used in an applied sports setting, these results provide comparative values for multiple force-time metrics within both eccentric and concentric phases of the CMJ movement that may help with sport-specific talent identification and may help with the development and tracking of amateur athletes' progress who are trying to reach peak performance potential and eventually compete on the professional level of play.

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